

TATEVIK TER-HOVHANNISYAN

Education

PhD student: Mathematical and software support for computing systems, complexes and computer networks 09/2026 - present
Russian-Armenian University, Yerevan, Armenia

Master of Science, with honors: Applied Mathematics and Informatics, AI and ML 09/2024 - 07/2026
Master's thesis: "*Knowledge Boundary Detection and Hallucination Mitigation via Uncertainty Calibration in Large Language Models*", GPA: 3.86 / 4
Russian-Armenian University, Yerevan, Armenia

Bachelor of Science: Applied Mathematics and Informatics 09/2020 - 07/2024
Bachelor's thesis: "Noisy text dataset generation for ASR output imitation", GPA: 3.14 / 4
Russian-Armenian University, Yerevan, Armenia

Specializations

AI Safety in LLM evaluations, 48h (2 ECTS) 03/2026 - 07/2026
Final project: "*Latent Alignment vs. Behavioral Alignment in Large Language Models*", in progress
Monoid.ru, online

Bioinformatics and Data Science in Oncology 09/2024 - 06/2025
Final project: "*Color Correction for Histology Imaging with Diffusion Models*"
Russian-Armenian University and BostonGene, Yerevan, Armenia

Publications and Conferences

- **NVIDIA** (2026). Nemotron 3 Ultra: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning. arXiv preprint arXiv:2606.15007.
- **Ter-Hovhannisyan, T.**, Aleksanyan, H., Avetisyan, K. (2024). *Adversarial attacks on language models: WordPiece filtration and ChatGPT synonyms*. Journal of Mathematical Sciences, Vol. 285, pp. 210–220 (2024). <https://doi.org/10.1007/s10958-024-07427-z>
 - Presented at the Artificial Intelligence and Natural Language Conference (AINL) 2023 in the Workshop on Artificial Intelligence Trustworthiness (WAIT), Yerevan, Armenia, April 21, 2023.
- **Ter-Hovhannisyan, T.**, Avetisyan, K. (2022). *Transformer-based multilingual language models in cross-lingual plagiarism detection*. In Proceedings of the Ivannikov Memorial Workshop (IVMEM) 2022, pp. 72-80. <https://doi.org/10.1109/IVMEM57067.2022.9983968>
 - Presented at Ivannikov Memorial Workshop (IVMEM) during Kazan Digital Week, Kazan, Russian Federation, September 23, 2022.

Experience

Applied Research Intern | NVIDIA 02/2025 - 07/2026

- Researching abstention, rejection behavior, and uncertainty-aware alignment in LLMs
- Evaluating robustness and safety properties of LLMs under SFT and RL training regimes

- Building an advanced scientific reasoning dataset in STEM, including synthetic data creation and analysis of the data quality and relevance using modern NLP techniques for Nemotron-3.

Research Advisor | TrainForge

09/2025 - 06/2026

- Helping my husband to start a startup in agentic safety.
- Participation in Plug-and-Play Armenia 2nd Batch incubation program.

Co-organiser and Manager | 1702

summers of 2021, 2025

- Managed the organisational processes for the non-profit's annual Summer School on programming for newly admitted and high school students. Conducted data collection to enhance the programme and organisation of the 1702 Summer School.

Bioinformatician Intern | Armenian Bioinformatics Institute

10/2024 - 11/2025

- Differentiated resistant and antibiotic-sensitive bacterial strains based on analysis of 5' endpoints distribution in the RNA degradome.

ML Engineer and Researcher | Center of Advanced Software Technologies

09/2021 - 05/2024

- Led the research and development of an adversarial attack research project on multilingual transformer-based models based on the token limitation of the models.
- Participated in a voice technology project and developed the product's prototype.
- Worked with the linguists' team on crowdsourcing for data annotations and refined data collection.

Teaching Experience

National Polytechnic University of Armenia

- **Corpus Linguistics Practice Lab Lecturer** 09/2026 - present
Developed and taught for 2nd-year bachelor's students in the Computational Linguistics program.

Russian-Armenian University

- **Intelligent Systems and AI Agents Lecturer** 09/2026 - present
Developed and taught a course on the modern state of AI systems for master's students in ML.
- **Assembly Language Practice Lab Lecturer** 02/2026 - present
Conducted seminars on Assembly language (SASM, NASM x86) and computer architecture for the 1st-year bachelor's students in the Applied Mathematics and Informatics program.
- **Data Analysis in Linguistics Teaching Assistant** 09/2022 - 12/2022
Conducted seminars and lectures for 2nd-year bachelor's students in the Linguistics program.

Russian School of Programming & MegaFon

- **Teaching Assistant** 03/2021 - 06/2021
Taught Data Analytics, Data Science, and AI to diverse groups, including middle school and Deaf children, in the 8-week program "AI: Quick Start in the Profession of the Future."

Ongoing Research Projects

- **Projection-Transport Steering: Conditional Control of Reasoning Behavior in LLMs**
Collaborative research project started during the SMILES 2026 Summer School on studying latent behavioral features to control LLM reasoning. *Future conference paper.*
- **Latent Alignment vs. Behavioral Alignment in LLMs**
Collaborative research project with Sabrina Sadiekh and participants from the Monoid AI Safety LLM Evals course. Investigating discrepancies between latent representations and observable behavior in LLMs, with a focus on mechanistic interpretability and safety evaluation. *Future conference paper.*

- **Discourse Connectives in the Armenian Language**

Collaborative research project with the Yerevan State University linguistics team. Contributing to the collection, annotation, and linguistic analysis of Armenian discourse connectives for computational and linguistic research applications. *Future applied research lab.*

- **Knowledge Boundary Detection and Hallucination Mitigation via Uncertainty Calibration in LLMs**

An extended master's thesis research project focused on uncertainty estimation, hallucination mitigation, and factual reliability in LLMs. Exploring calibration methods for detecting knowledge boundaries and improving abstention behavior under uncertain conditions. *Future conference paper.*

Courses, Summer Schools, Awards & Certifications

- International Youth Neuroscience Association (IYNA) Virtual Summer School, 2026 – *Neural biology*
- NeuroSymbolic AI Summer School, 2026, 2025 – *Neuro-Symbolic approaches and research*
- SMILES Summer School in Suzhou, China, 2026, **Best Poster Award (1st place), Ascend special nomination for "Latent Behavioural Features for Controlling LLM Reasoning" research project** – *Uncertainty quantification, Bayesian ML, LLMs for low-resource languages*
- UniDive 2nd Training School in Yerevan, 2026 – *Diversity Quantification, Linguistic Typology for NLP*
- LLM4ETL Hackathon, 2025, **Special Award** – *ETL with LLMs solution for call centers*
- Krisp AI Hackathon, 2025, **2nd place** – *Efficient RAG system development*
- Educational Award of the President of the Republic of Armenia in the IT Sphere, 2025, **Best Master Student II Category**
- Armenian LLM Summer School, 2025, 2024 – *LLMs, RAG, Interpretability, LLM Security*
- Eghiazaryan Cup Programming Contest, 2026, **20th place**; 2025, **11th place**; 2022, **11th place**
- Yerevan Academy for Linguistics and Philosophy (YALP) Summer School, 2025, 2024, 2023 – *Phonetics, Neurolinguistics, Writing Systems, Semantics, Bilingualism*
- Deep Origin DO Challenge, 2025, **1st place** – *Molecular modelling of protein and ligand relations with ML and DL methods*
- Hackathon by Ministry of HTI, 2024, **3rd place** – *IoT solution for agritech*
- LST Summer School, 2024 – *Psycho- and Neurolinguistics, Phonetics and Speech Technology, NLP*
- Yandex School of Data Analysis (YSDA), 2023-2024, as part of RAU course – *Intro to Linguistics, Python, ML1, ML2*
- Sirius Educational Center, 2021 – *Phonetics and Writing, Morphology, Anthropology*
- RAU Programming Winter School, 2021 – *Competitive programming, Algorithms, C++*
- PCAP Programming Essentials in Python, 2021, **PCAP Certificate**
- Deep Learning School, 2018-2019, **2nd degree diploma** (1st semester), **1st degree diploma (2nd semester)** – *Computer Vision, NLP, Audio Processing*

Skills

ML, DL, Multilingual NLP, Voice Technologies, LLMs, SFT, RL, RAG, Mechanistic Interpretability, AI Robustness, LLM Security, Fairness & Trustworthy AI, LLM Robustness, Alignment and Evaluation, TensorFlow, PyTorch, PyTorch-Lightning, W&B, Hugging Face, Dataset Design & Evaluation, Prompt Engineering, MLFlow, NemoSkills, Nemo, Python, C++, JavaScript, Bash, Docker, Git, Linux, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, REST API, REACT, Scikit-Learn, SpaCy, Slurm, Statistical analysis, Data curation & annotation, Mentoring interns, educational content development, RNA-seq/DNA analysis, Diffusion Models for biomedical imaging, Discourse and Dialogue Structure Analysis, Multilingual Tokenization, WordPiece Analysis, Phonetics and Phonology Analysis, Praat, Language Typology

Research interests

Mechanistic interpretability, LLM security, uncertainty estimation, hallucination mitigation, multilingual NLP, low-resource language technologies, alignment, and trustworthy AI.

Languages

Russian (native), Armenian (native), English (proficient), French (intermediate), Turkish (beginner)